

PEDAMALLU SAI MRUDULA

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EDUCATION

B.Tech in Amrita Vishwa Vidyapeetham

July 2024 – July 2028

Artificial Intelligence & Computer Science (CGPA:8.51)

Coimbatore, India

PROJECTS

Smart Grid Intrusion Detection (IEEE Access) | [Python](#), [ML](#), [Cybersecurity](#)

2025

- First-author *IEEE Access* publication proposing a privacy-aware intrusion detection framework for smart grid communication under severe class imbalance.
- Built a hybrid anomaly detection and classification pipeline with optimized feature selection (SHAP, RFE, PCA), evaluated on real IEC 60870-5-104 SCADA traffic.
- Achieved 99.29% accuracy, 94.8% recall, and 4.1% false positive rate, with robustness validated against unseen attacks.

Real-Time Markerless Multimodal Motion Capture | [MediaPipe](#), [PyTorch](#) [CUDA](#), [ZMQ](#) 2025–2026

- Developed a real-time dual-camera stereo motion capture system achieving 25–30 FPS with $\pm 1-2$ cm 3D reconstruction accuracy, representing a $5\times$ improvement over a single-camera baseline through GPU-accelerated triangulation (5 ms/frame) and network-synchronized frame matching (≈ 100 ms latency) using a ZMQ pub/sub architecture across dual laptops.
- Designed an 8-stage processing pipeline generating 33-landmark 3D human pose estimations with 18+ derived joint angles, velocities, and accelerations, integrating adaptive 1-Euro filtering and ArUco-based camera calibration (≈ 1 px reprojection error) while storing structured biomechanical data in SQLite via asynchronous batch writes for reproducible offline analysis.

AI Microservices Architecture (RAG-Based NLP Services) | [FastAPI](#), [LangChain](#)

2025

- Designed modular RESTful microservices for document summarization and question answering using transformer models.
- Built a retrieval-augmented generation (RAG) pipeline with embedding-based vector search, optimizing service isolation, scalability, and latency-aware inference for modular, independent deployment of NLP components.

MisSpoke – Real-Time Spoken Language Intelligence System | [Speech Processing](#), [NLP](#)

2025

- Designed a real-time spoken language intelligence system to model fluency, filler usage, and speaker confidence using low-latency speech-to-text and NLP pipelines.
- Implemented adaptive, human-centered feedback loops that dynamically adjust conversational guidance based on linguistic patterns and interaction context.

EEG Signal Reconstruction with Local Graph Learning (ADMM) | [GSP](#), [ADMM](#)

2025

- Developed a graph-based EEG signal reconstruction framework using the Alternating Direction Method of Multipliers (ADMM) to recover corrupted neural signals under 10–90% missing data while maintaining stable reconstruction error (RMSE ~ 0.35).
- Reduced reconstruction error by **65%** compared to a zero-filling baseline (RMSE $\sim 1.0 \rightarrow 0.35$) and preserved signal fidelity with **SNR up to 9.34 dB** even under extreme 90% data loss.
- Designed a joint graph-learning and signal recovery pipeline leveraging local graph smoothness across EEG channels, enabling accurate reconstruction of neural activity patterns and outperforming baseline methods across RMSE, NMSE, and SNR metrics.

TECHNICAL SKILLS

Languages: Python, HTML/CSS/JavaScript, SQL, Supabase

ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, OpenCV

Data & Analysis: NumPy, Pandas, Matplotlib

Frameworks & Tools: LangChain, Rasa, Agora, Git, Wireshark

EXPERIENCE

Academy of Robotics Pvt. Ltd. [🔗](#)

Jul 2024 – Sep 2024

AI Chatbots Intern

Visakhapatnam, India

- * Gained hands-on experience with the Rasa framework by building and configuring basic conversational chatbots, understanding intent classification, entities, and dialogue flows.
- * Applied prompt engineering techniques to design structured and interactive chatbot conversations, including simple conversational games and user-driven interactions.

LEADERSHIP & VOLUNTEERING

Amrita Vishwa Vidyapeetham

2024 – Present

- Co-founder and Vice President of Init Club, a student-led technical initiative enabling cross-department collaboration and open-source projects; led all non-technical operations including coordination, outreach, and event execution.
- Active member of **IETE (AI/ML)**, **NSS**, and **Poetree Club**, contributing to technical workshops and campus initiatives; served as **Class Representative** for one academic year.

CERTIFICATIONS & ACHIEVEMENTS

- 4th Place – Agora GDG Hackathon 
- NASA Space Settlement Contest 2022 
- Publication in *Energy Conversion and Management: X* 